Lec 5

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Relational Model Constraints (cont’d.)

- 1. Schema-based constraints (explicit constraints): Rules that can be defined directly in the database schema, such as a student's ID must be unique.
- 2. Application-based constraints (semantic constraints or business rules): Rules that cannot be defined directly in the database schema and must be checked by an application program, such as an employee's bonus must not exceed a specific limit.

Domain Constraints

- Typically include:
  - Numeric data types for integers and real numbers
  - Characters
  - Booleans
  - Fixed-length strings
  - Variable-length strings
  - Date, time, timestamp
  - Money
  - Other special data types

Key Constraints and Constraints on NULL Values

- 1. Unique tuples
- No two rows in a table can have exactly the same values for all their attributes.
- Example: Two students cannot have identical values in every column.
- Superkey
- A superkey is one or more attributes that uniquely identify each row in a table.
- Example: In a Student table, Student_ID is a superkey because it identifies each student uniquely. {Student_ID, Name} is also a superkey.
- Key (Candidate Key)
- A key is a minimal superkey. It uniquely identifies each row, and removing any attribute means it can no longer identify rows uniquely.
- - Student_ID is a key and a superkey.
- - {Student_ID, Name} is a superkey, but not a key, because Name is unnecessary.
- In short: Every key is a superkey, but not every superkey is a key.

Key Constraints and Constraints on NULL Values (cont’d.)

- 1. Uniqueness: No two rows in a table can have the same value for the key attributes.
- 2. Minimality: A key contains no unnecessary attributes. If we remove any attribute, it will no longer uniquely identify each row.
- Example: In a Student table, Student_ID is a key because every student has a unique ID, and no extra attribute is needed to identify the student.

Key Constraints and Constraints on NULL Values (cont'd.)

- **Candidate key**
  - Relation schema may have more than one key
- **Primary key** of the relation
  - Designated among candidate keys
  - Underline attribute
- Other candidate keys are designated as **unique keys**

Key Constraints and Constraints on NULL Values (cont'd.)

CAR

License_number	Engine_serial_number	Make	Model	Year
Texas ABC-739	A69352	Ford	Mustang	02
Florida TVP-347	B43696	Oldsmobile	Cutlass	05
New York MPO-22	X83554	Oldsmobile	Delta	01
California 432-TFY	C43742	Mercedes	190-D	99
California RSK-629	Y82935	Toyota	Camry	04
Texas RSK-629	U028365	Jaguar	XJS	04

Figure 3.4
The CAR relation, with two candidate keys: License_number and Engine_serial_number.

Relational Databases and Relational Database Schemas

- **Relational database schema** *S*
  - Set of relation schemas $S = \{R_1, R_2, \ldots, R_m\}$
  - Set of integrity constraints IC
- **Relational database state**
  - Set of relation states $DB = \{r_1, r_2, \ldots, r_m\}$
  - Each r_i is a state of R_i and such that the r_i relation states satisfy integrity constraints specified in IC
  - DB is the complete database, containing a set of relations (tables): (DB={r_1,r_2,r_m}).
  - Each (r_i) represents the current data (rows) stored in a relation (R_i).
  - IC (Integrity Constraints) are rules that the data must follow to keep the database correct and valid.

Relational Databases and Relational Database Schemas (cont'd.)

- **Invalid state**
  - Does not obey all the integrity constraints
- **Valid state**
  - Satisfies all the constraints in the defined set of integrity constraints IC

Integrity, Referential Integrity, and Foreign Keys

- **Entity integrity constraint**
  - No primary key value can be NULL
- **Referential integrity constraint**
  - Specified between two relations
  - Maintains consistency among tuples in two relations

Integrity, Referential Integrity, and Foreign Keys (cont'd.)

- **Foreign key** rules:
  - The attributes in FK have the same domain(s) as the primary key attributes PK
  - Value of FK in a tuple t_1 of the current state $r_1(R_1)$ either occurs as a value of PK for some tuple t_2 in the current state $r_2(R_2)$ or is NULL

Integrity, Referential Integrity, and Foreign Keys (cont'd.)

- Diagrammatically display referential integrity constraints
  - Directed arc from each foreign key to the relation it references
- All integrity constraints should be specified on relational database schema

Other Types of Constraints

- Semantic integrity constraints: are business rules that ensure data follows the organization's requirements.
- - Triggers: Database rules that automatically execute when an event occurs, such as inserting or updating data.
- - Assertions: Rules that specify conditions the database must satisfy must always be satisfied by the database to maintain data correctness.
- . However, many DBMSs do not support assertions directly.
- that Example: A company's total employee salaries must not exceed its budget.
- An assertion ensures that the database always follows a specified rule.

Other Types of Constraints

- - Application programs: These rules are often checked by the application before data is saved.
- Example: A university requires that a student register for no more than 6 courses per semester. The application can check this rule before accepting registration.

Other Types of Constraints (cont’d.)

- **Functional Dependency (FD)**
- **A functional dependency is a relationship between two sets of attributes in a database table, where the value of one attribute (X) determines exactly one value of another attribute (Y).**
- **This means X determines Y.**
- **Example: Student table**
- **Student_ID (X) Student_Name (Y)**
- **101 Ahmed**
- **102 Sara**
- **103 Ali**
- **101 Ahmed**

Other Types of Constraints (cont’d.)

- **Step-by-step explanation:**
- **1. Student_ID is the attribute X.**
- **2. Student_Name is the attribute Y.**
- **3. If we know Student_ID = 101, we can determine that the student's name is Ahmed.**
- **4. The same Student_ID cannot be associated with two different student names in a valid instance of this functional dependency**
- **State constraints**
  - Define the constraints that a valid state of the database must satisfy
- **Transition constraints**
  - Define to deal with state changes in the database

Update Operations, Transactions, and Dealing with Constraint Violations

- Operations of the relational model can be categorized into retrievals and updates
- Basic operations that change the states of relations in the database:
  - Insert
  - Delete
  - Update (or Modify)

Figure 3.6
One possible database state for the COMPANY relational database schema.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Figure 3.6
One possible database state for the COMPANY relational database schema.

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

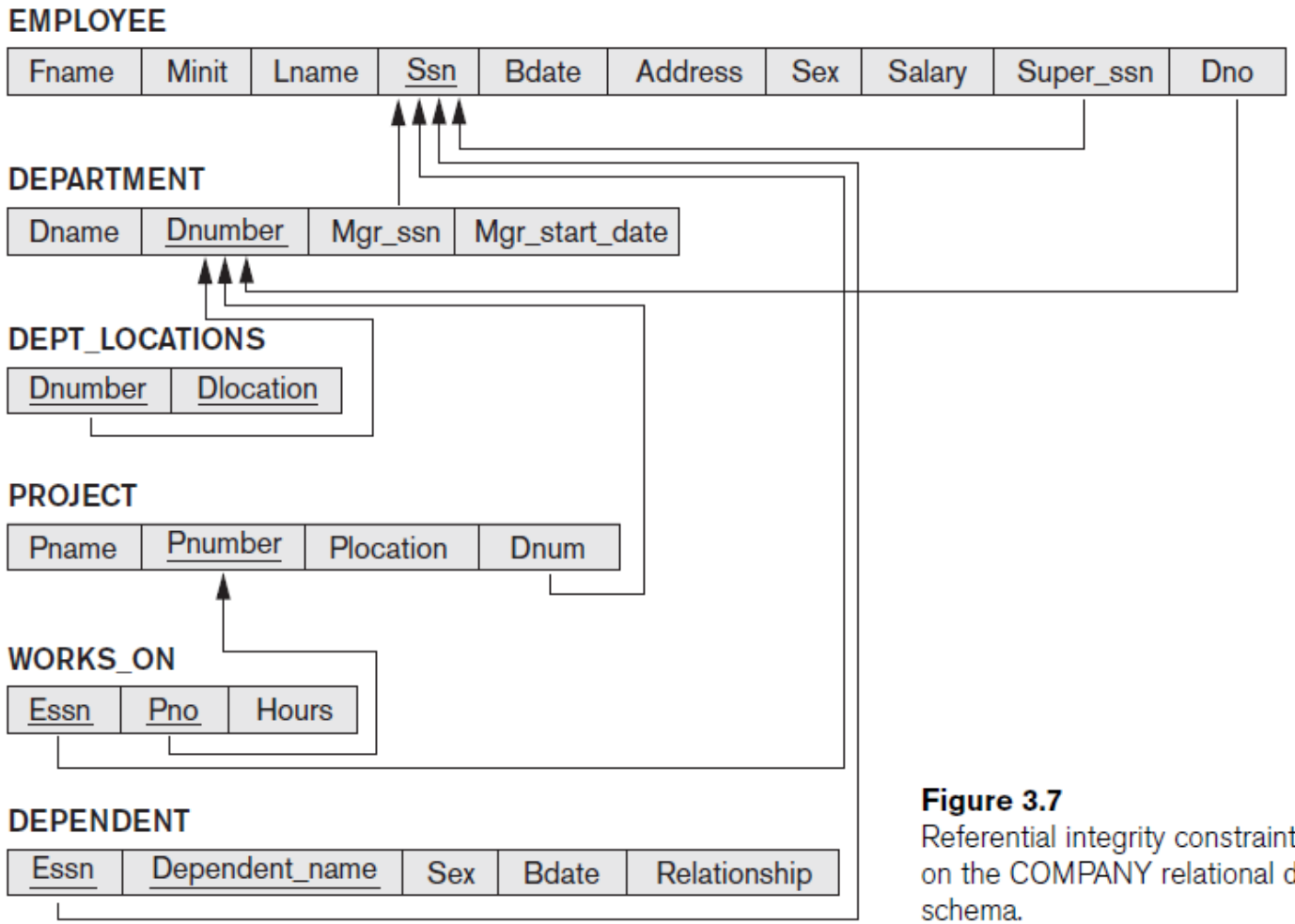


Figure 3.7
Referential integrity constraints displayed on the COMPANY relational database schema.

The Insert Operation

- Provides a list of attribute values for a new tuple t that is to be inserted into a relation R
- Can violate كسر any of the four types of constraints
- If an insertion violates one or more constraints
  - Default option is to reject the insertion

The Delete Operation

- Can violate only referential integrity
  - If tuple being deleted is referenced by foreign keys from other tuples
  - **Restrict**
    - Reject the deletion
  - **Cascade**
    - Propagate the deletion by deleting tuples that reference the tuple that is being deleted
  - **Set null** or **set default**
    - Modify the referencing attribute values that cause the violation

The Update Operation

- Necessary to specify a condition on attributes of relation
  - Select the tuple (or tuples) to be modified
- If attribute not part of a primary key nor of a foreign key
  - Usually causes no problems
- Updating a primary/foreign key
  - Similar issues as with Insert/Delete

The Transaction Concept

- **Transaction**
  - Executing program
  - Includes some database operations
  - Must leave the database in a valid or consistent state
- **Online transaction processing (OLTP) systems**
  - Execute transactions at rates that reach several hundred per second